

2015年3月18日

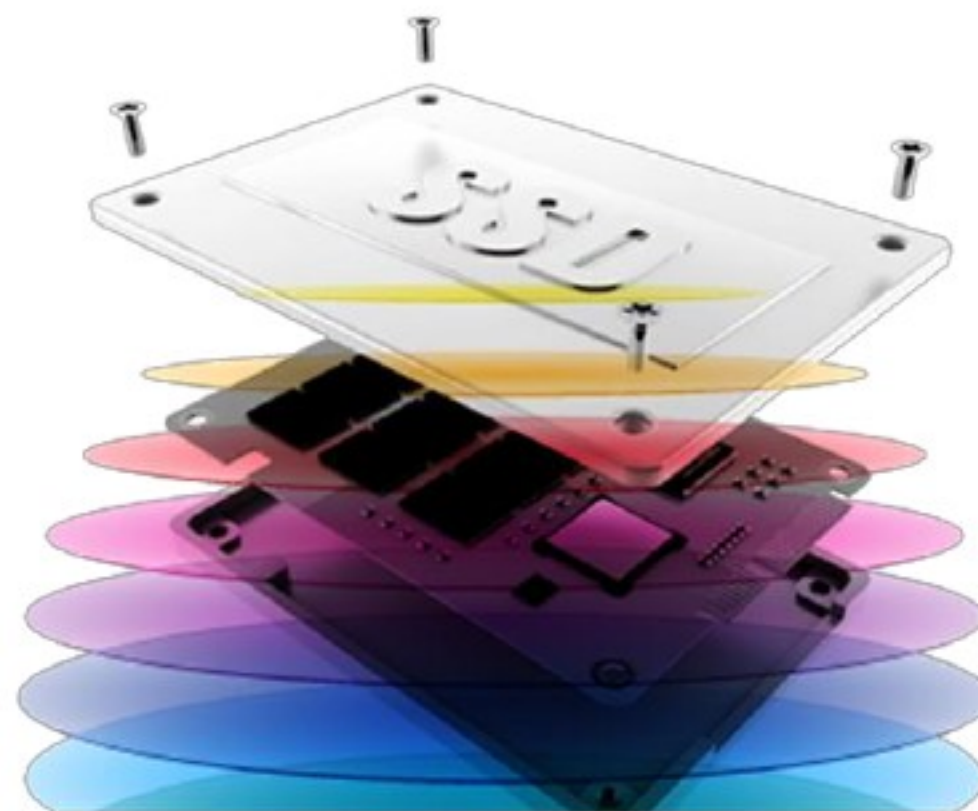
TESLA MODEL S TEARDOWN

IHS Technology

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IHS Teardown Service Brief

Who need the teardown services?

Component Manufacturers

- ICs
- Passive components
- Others

OEMS

- Handsets/tablets
- PCs/Servers
- Automotive electronics
- Industrial equip.
- Other

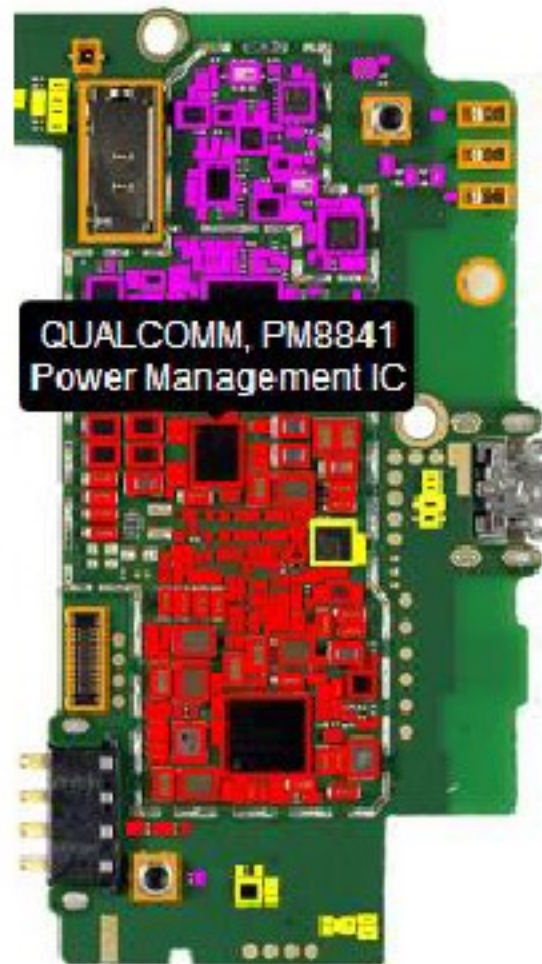
End Users / Other

- Service providers
 - Wireless
 - Cable/satellite
 - Utilities
- Retailers
- OEMS that are using ODMs to build for them
- Financial analysts

Main Customers

An IHS Teardown is...



Bill-of-Materials (BoM) Created from Hardware; Primary focus is “Should Cost” hardware cost assessment

Based on assumptions (Volume, country of origin, lifetime, EMS Providers, etc.) and snapshot in time

Leverages knowledge base (products, people, and practice areas) at IHS

Disassembly / Photography / Die Photos and Comprehensive identification From ICs to 0201 passives and screws (Manufacturers and PN#s Identified where possible)

An IHS Teardown is NOT...

BOM Analysis from client-provided BOM

**Reverse engineering / Schematics /
Performance testing**

**Perfect knowledge / Customer's precise
costs**

**A forecast (CPT and other services can
provide)**

**Cost of IP, Royalties, R&D, Or Post-Mfg
Costs**

Teardown Team and Research Analysts

IHS Globally ~ 200 employees

Teardown Service = ~ 30 employees

United States

- El Segundo, CA - 10 Teardown analysts, Wireless Team, Memory Team (DRAM), Component Price Tracker Team
- Austin, TX – 2 Teardown analysts
- Santa Clara - Display & Touchscreen Team, Memory Team (NAND), Semiconductor Research (IC Modeling)

APAC

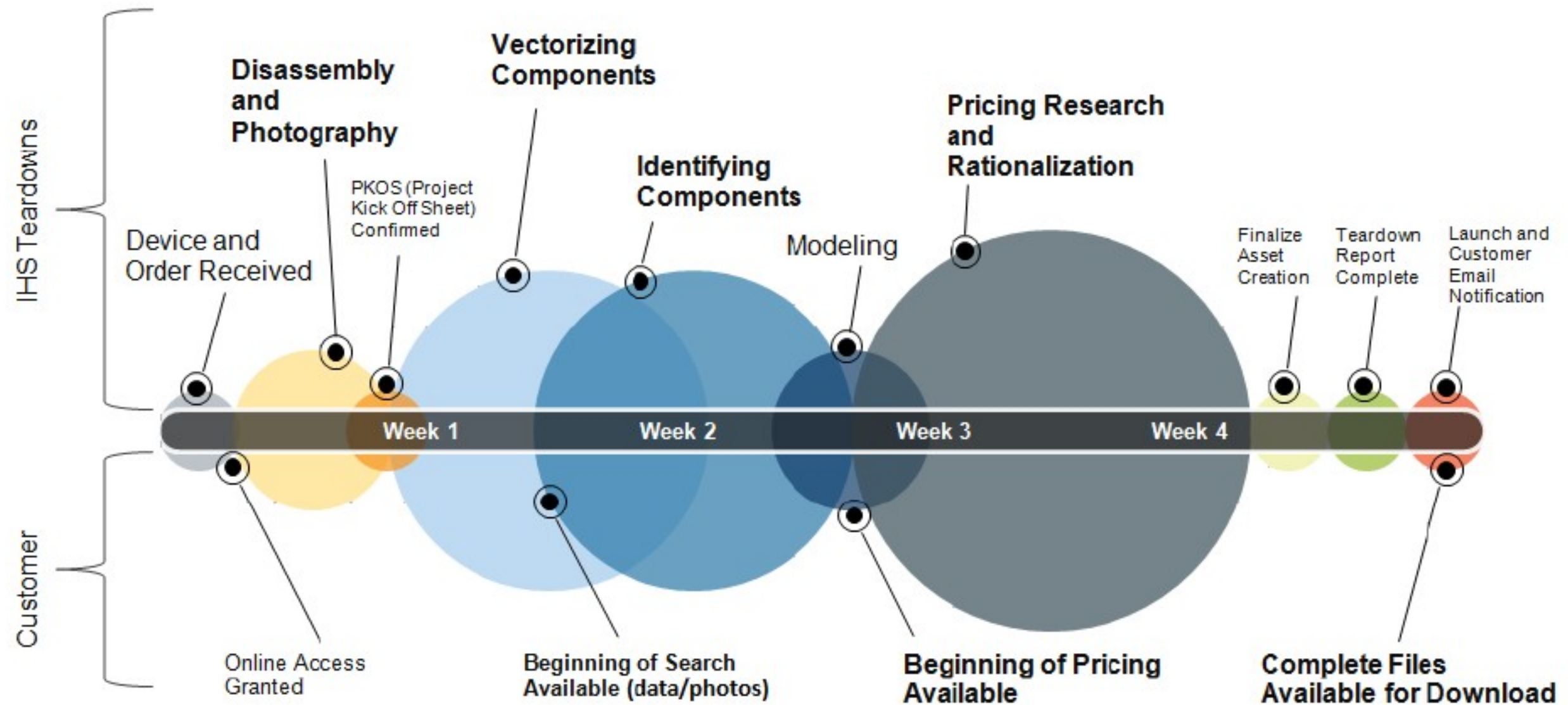
- Shenzhen - 12 Teardown analysts, 2 Dedicated Pricing Researchers
- Seoul – 2 Teardown analysts, Display & Touchscreen Team

EMEA

- Munich - MEMS and Sensors Team
- India – 3 Teardown analysts



What is the typical teardown process?



MILESTONES MAGNITUDE IMPORTANCE



Less



More

Average

Greater

Tesla - 2013 Model S Premium Media Control Unit

IHS have done...



Premium Media Control Unit



Premium External Audio Amplifier



HVAC Control Unit



Thermal Controller



Instrument Cluster



Sunroof Control Unit



Liftgate Left Hand Taillight



Passive Safety Restraints Control Module



Charger



Body Control Unit



Power Liftgate Module

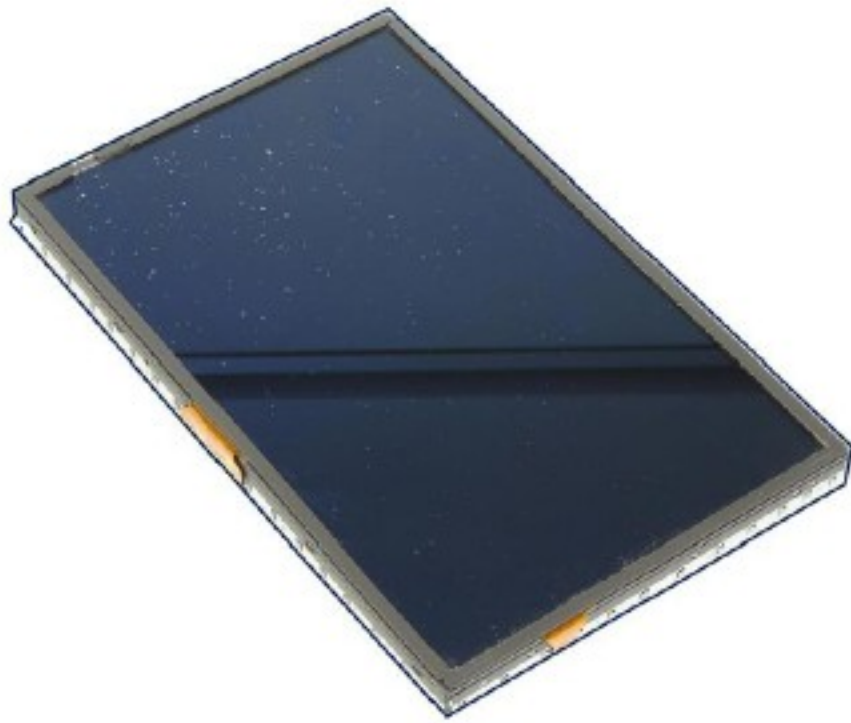


Park Brake Controller

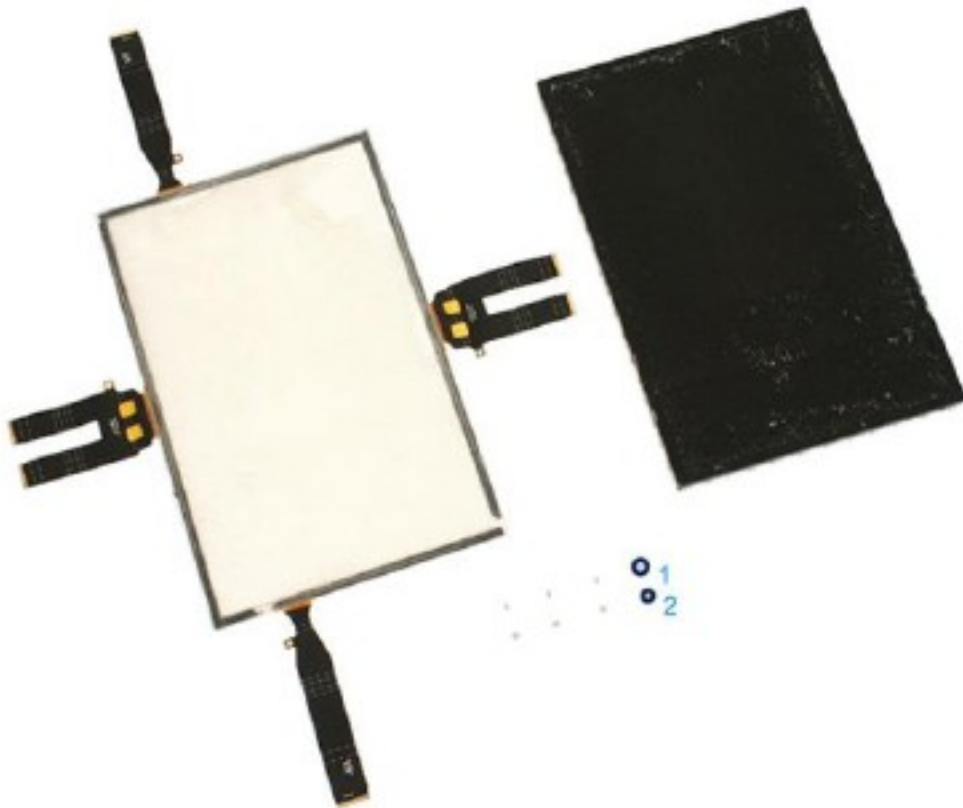
Tesla - 2013 Model S Premium Media Control Unit



Touchscreen Module

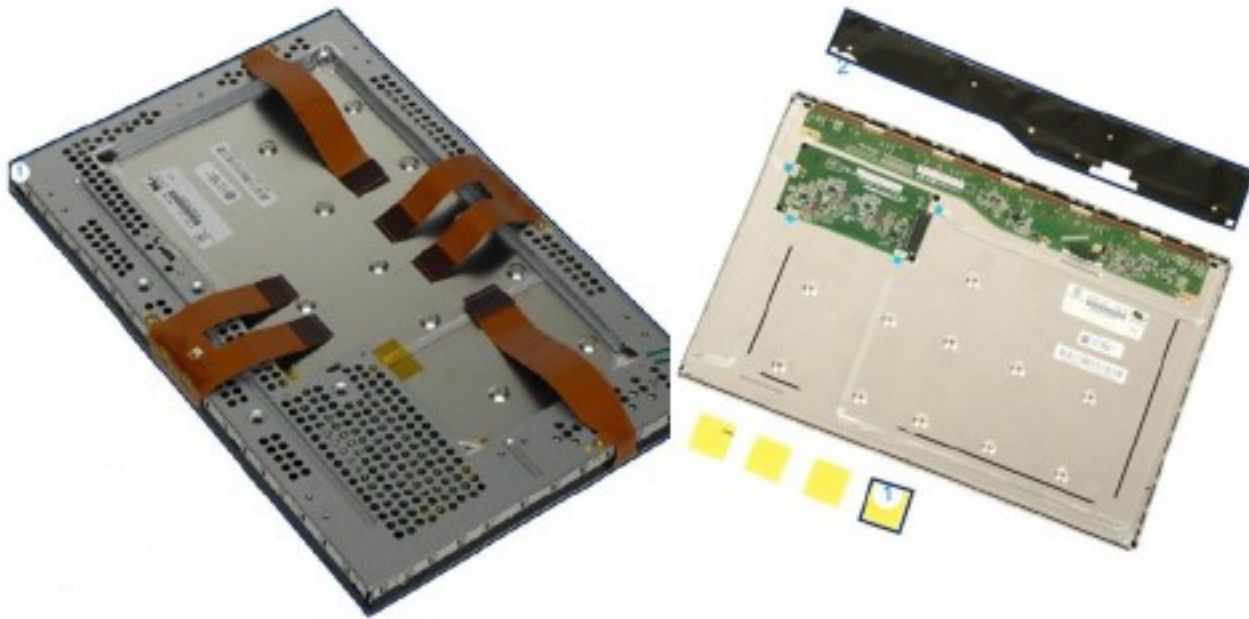


**\$85.00 - Display Window /
Touchscreen Assembly, 17"
Diagonal, Capacitive, GG Type
(SITO), w/ Cover Glass, w/ Integral
Flex PCB - MFG: TPK Holdings -
(Qty: 1)**

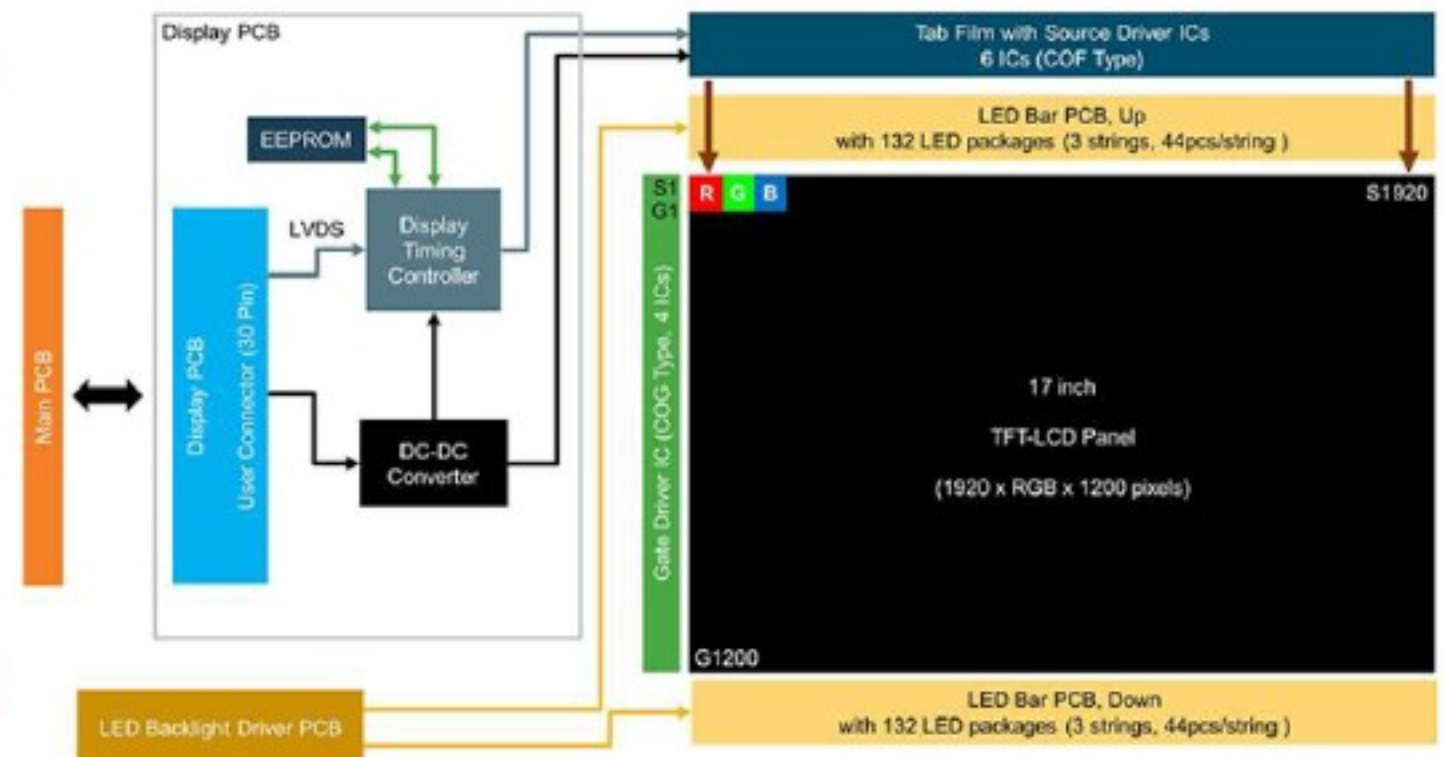


Source: BITKOM, Siemens

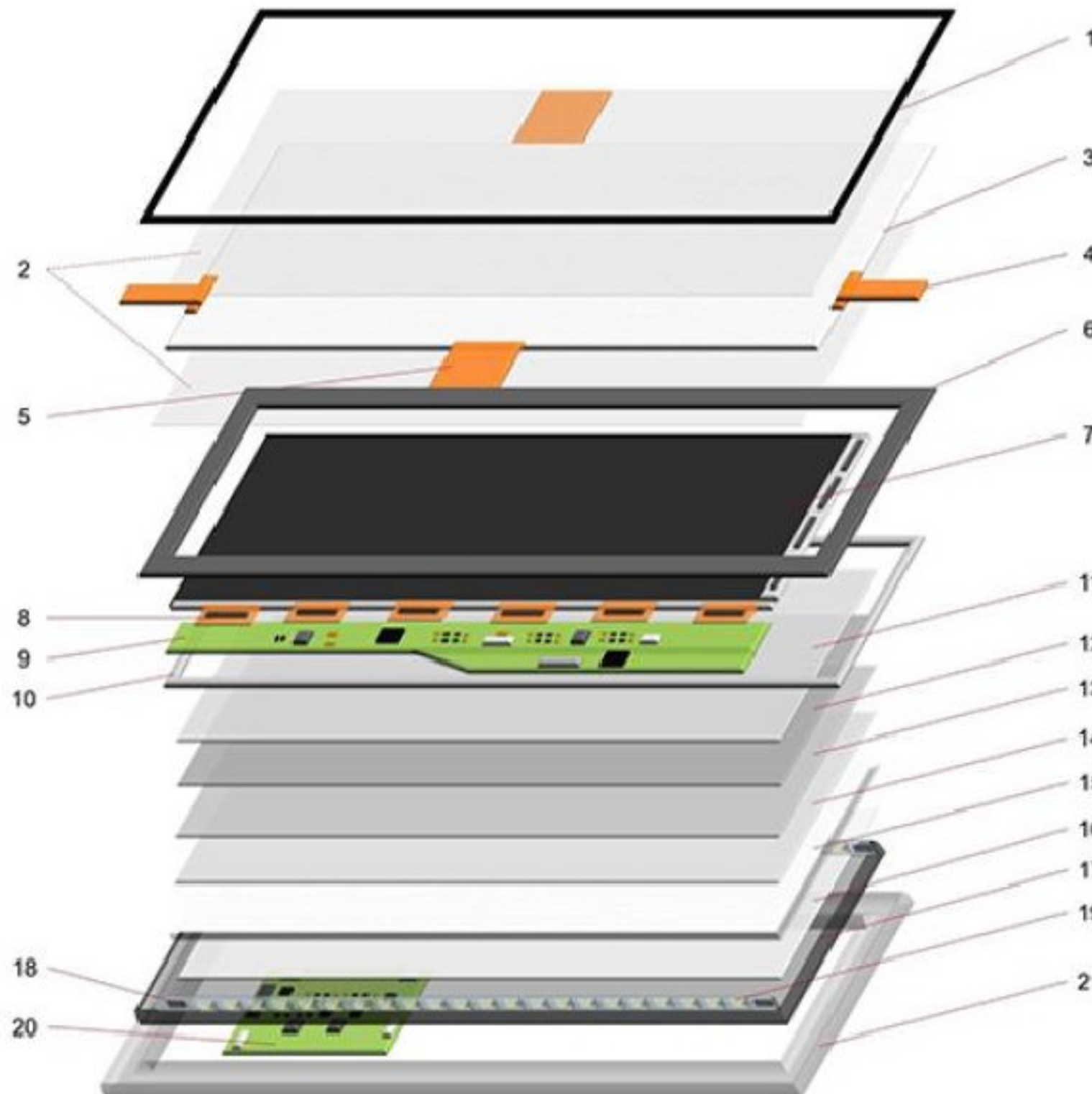
Display Module



\$180.00 - Display Module, 17"
Diagonal, 16.7M Colors TFT (MVA
Mode), 1920 x 1200 Pixels, 133 ppi,
1,740 g - MFG: INNOLUX CORP -
MPN: G170J1-LE1 - (Qty: 1)

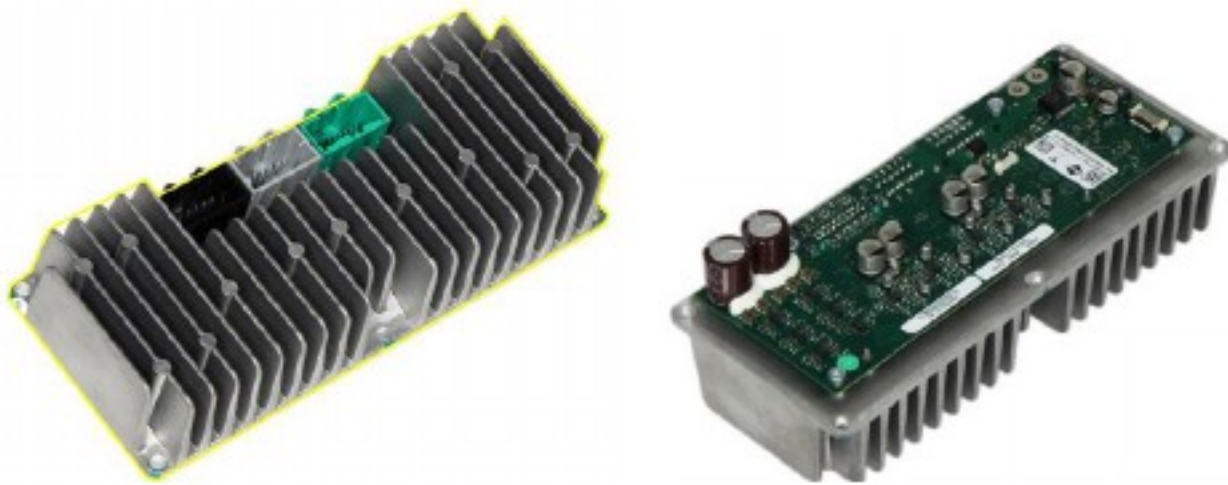


Display/Touchscreen Module - Overview



No	Component
1	Cover Glass
2	OCA
3	ITO Sensor Glass
4	Touchscreen PCB, Y
5	Touchscreen PCB, X
6	Display Top Frame
7	LCD Cell
8	TAB Film
9	Display PCB
10	Display Mold Frame
11	Diffusion Sheet (Upper)
12	Prism Sheet (Upper)
13	Prism Sheet (Lower)
14	Diffusion Sheet (Lower)
15	Light Guide Plate
16	Reflection Sheet
17	Display Bottom Frame
18	LED Bar PCB
19	LED Package
20	LED Backlight Driver PCB
21	Display Shield Cover

Audio Amplifier Module



\$65.00 - Audio Amplifier Module, 7-Channel, w/ Microchip 16-Bit MCU - MFG: S1NN GMBH - MPN: 1000321-002-11 - (Qty: 1)

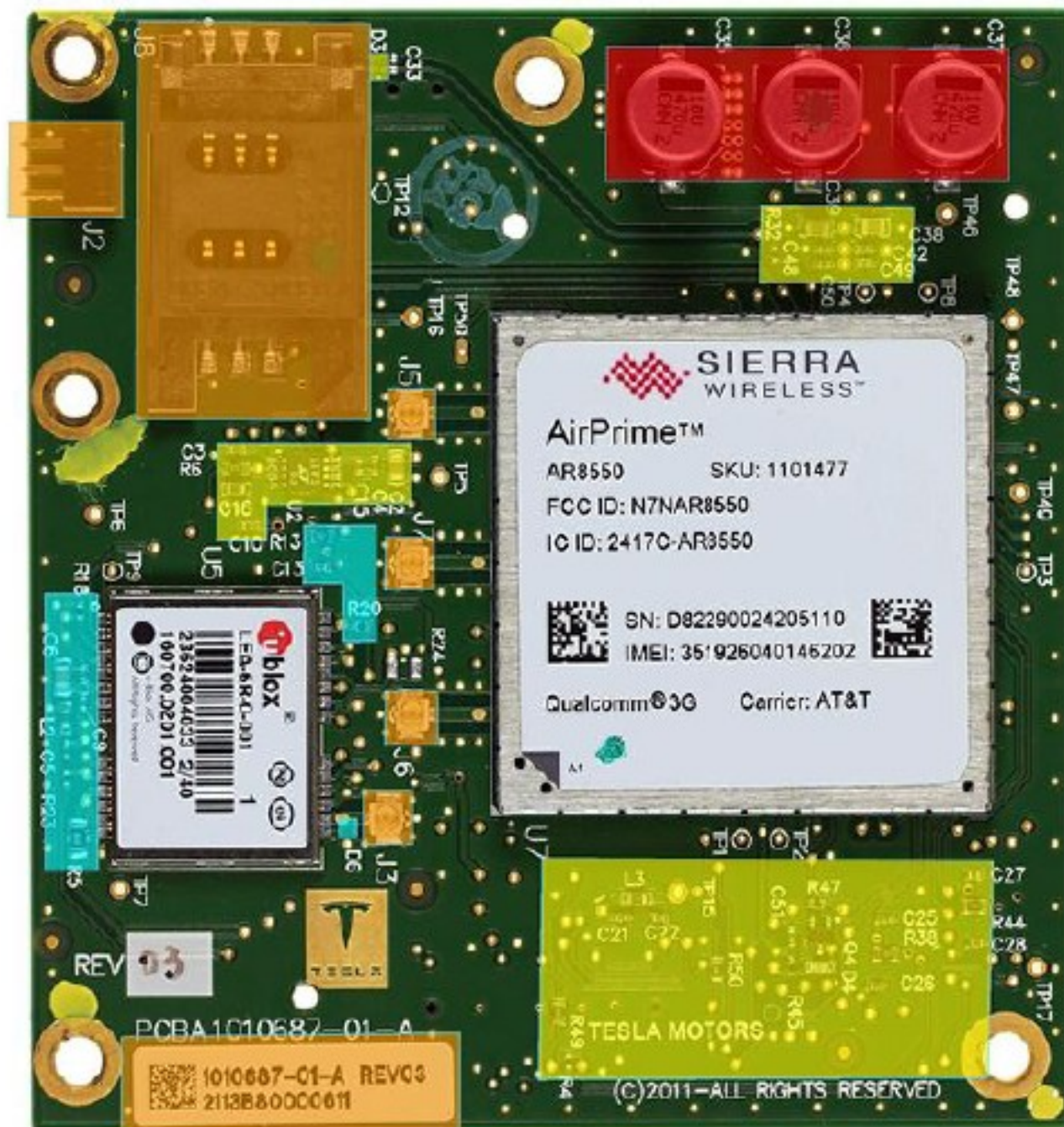
Audio Codec, 24-Bit, 8-Channel DAC, 192kHz, w/ PLL & Differential Outputs, Automotive

Microchip Technology , PIC18F25K22, MCU, 16-Bit, 32Mhz, 32KB Flash,



Communication

\$45.00 - Cellular Module, Quad-Band GSM/EDGE, Tri-Band WCDMA/HSPA, Automotive - MFG: SIERRA WIRELESS INC - MPN: AR8550 - (Qty: 1)



Functional Areas

- Power Supply
- I/O & Interface
- Mechanical / Electro-Mechanical
- Memory
- GPS

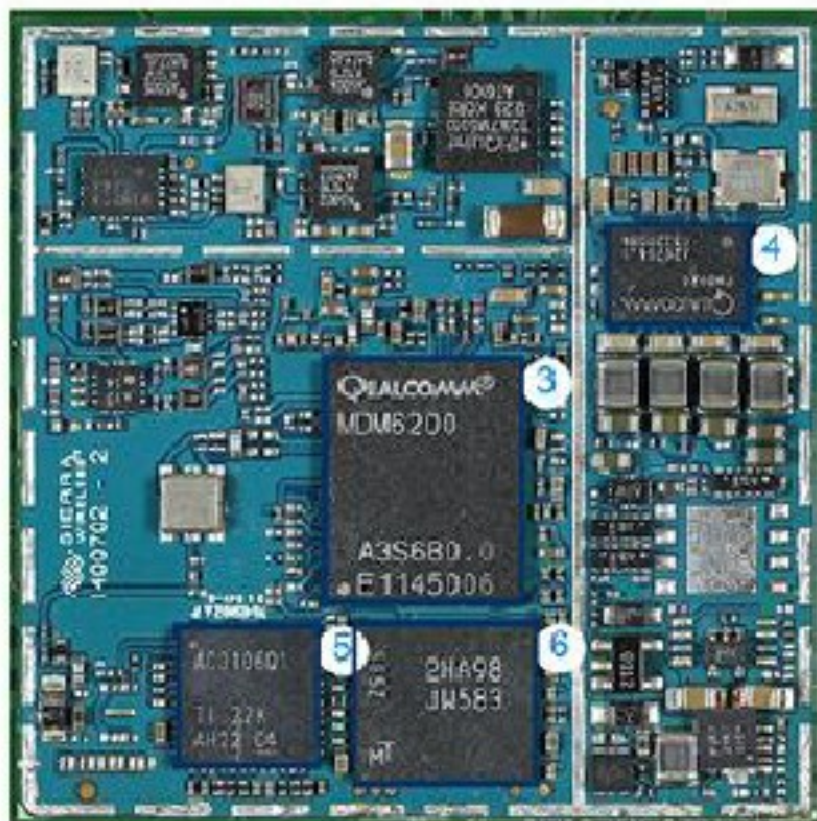
Components

- 1. LINEAR TECHNOLOGY CORP, LTC1860HMS8, ADC, 5V, 12-Bit, 1-Channel
- 2. SIM Card, 6 Gold Plated Contacts in Black Plastic Carrier, w/ Plastic Retainer Lock
- 3. SIM Card

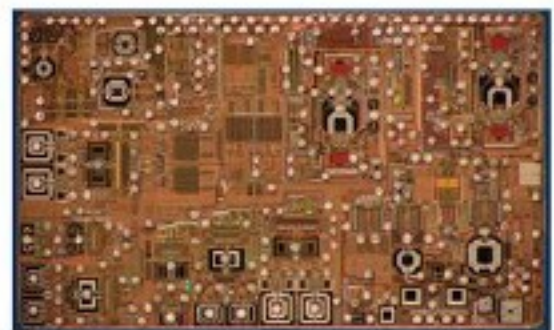
Communication PCB – Cellular Module

Components

1. SMT Cover, Stamped / Formed Metal
2. SMT Frame, Stamped / Formed Metal
3. QUALCOMM, MDM6200, Baseband / RF Transceiver, GSM/EDGE/HSPA+, 45nm
4. QUALCOMM, PM8028, Power Management IC
5. TEXAS INSTRUMENTS INC, 6PAIC3106IRGZRQ1, Audio Codec, Stereo, w/ Stereo Headphone Amplifier, Automotive
6. MICRON TECHNOLOGY INC, MT29C2G24MAAAHAMD-5 IT, MCP, 2Gb NAND Flash + 1Gb Mobile DDR, 200MHz

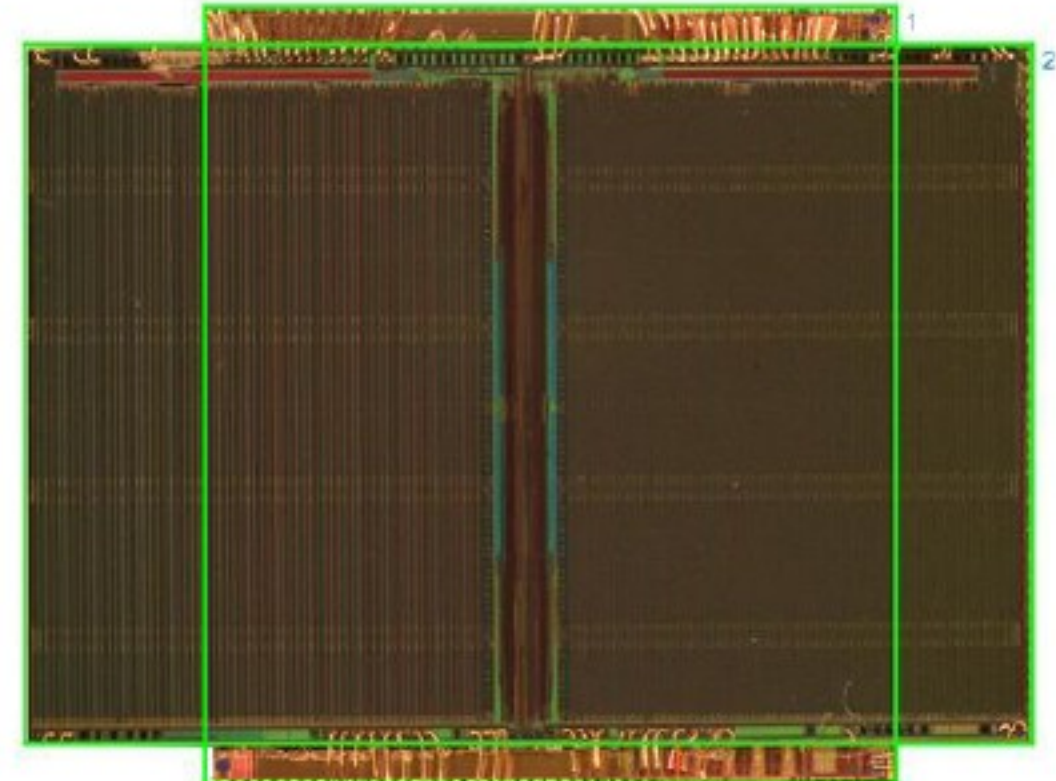


QUALCOMM Baseband



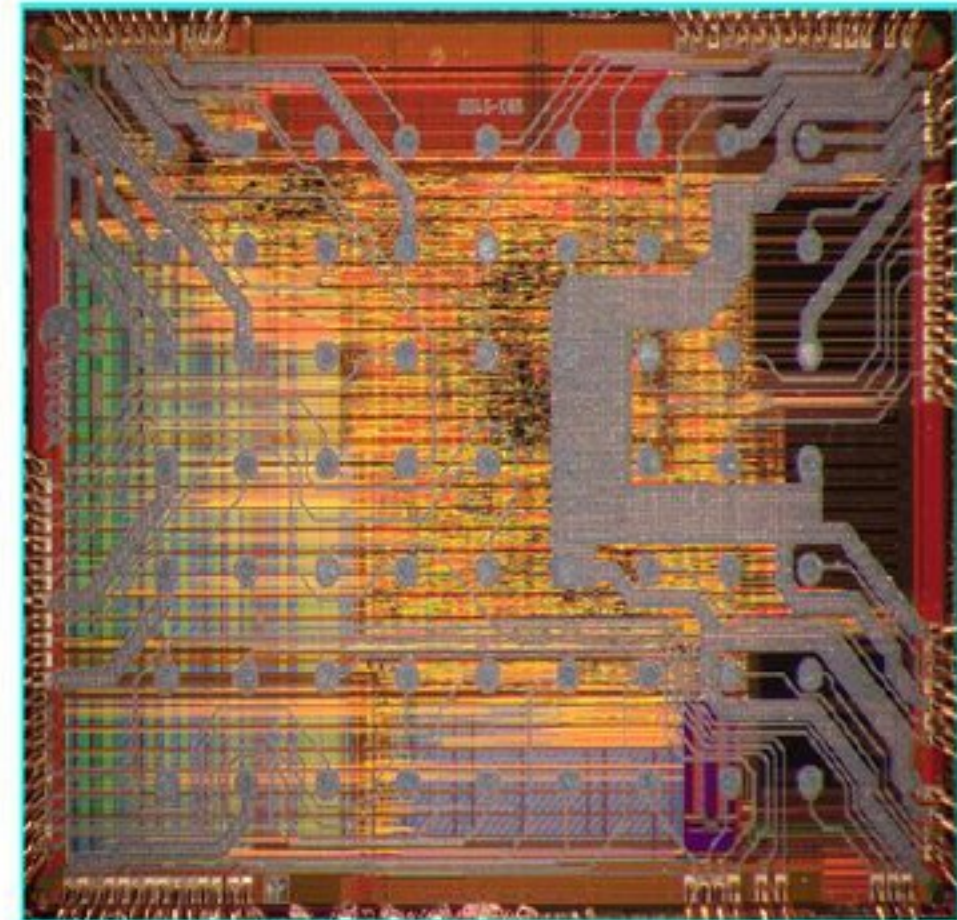
QUALCOMM RF Receiver

MICRON 2Gb NAND Flash



MICRON 1Gb Mobile DDR

Communication – GPS Module



Components

1. U-BLOX AG,UBX-G600-BT-B7300,GPS Baseband,ARM7TDMI-S CPU,GPS / SBAS Engine,3GPP Compliant,Automotive
2. MICROCHIP TECHNOLOGY INC,SST39VF400A,70-nb,MAOE,Flash,NOR,4Mb,70ns
3. U-BLOX AG,UBX-G0010-QA,GPS Receiver,w/ Integrated LNA,Automotive
4. SMT Cover,Stamped / Formed Metal

Components

1. U-BLOX AG,UBX-G600-BT-B7300,1c Die - GPS Baseband, ARM7TDMI-S CPU, GPS / SBAS Engine, 3GPP Compliant, Automotive

Functional Areas

- GPS



Main PCB

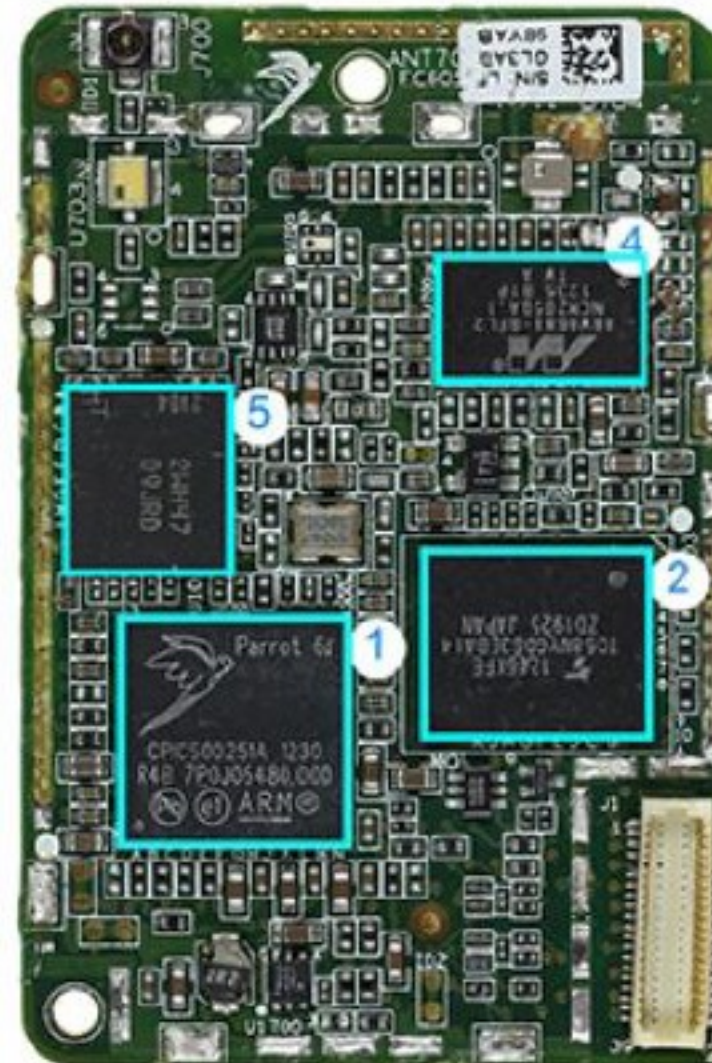


Functional Areas

- Mechanical / Electro-Mechanical
- Memory
- Glue Logic
- MCU
- Processing
- Power Supply
- I/O & Interface



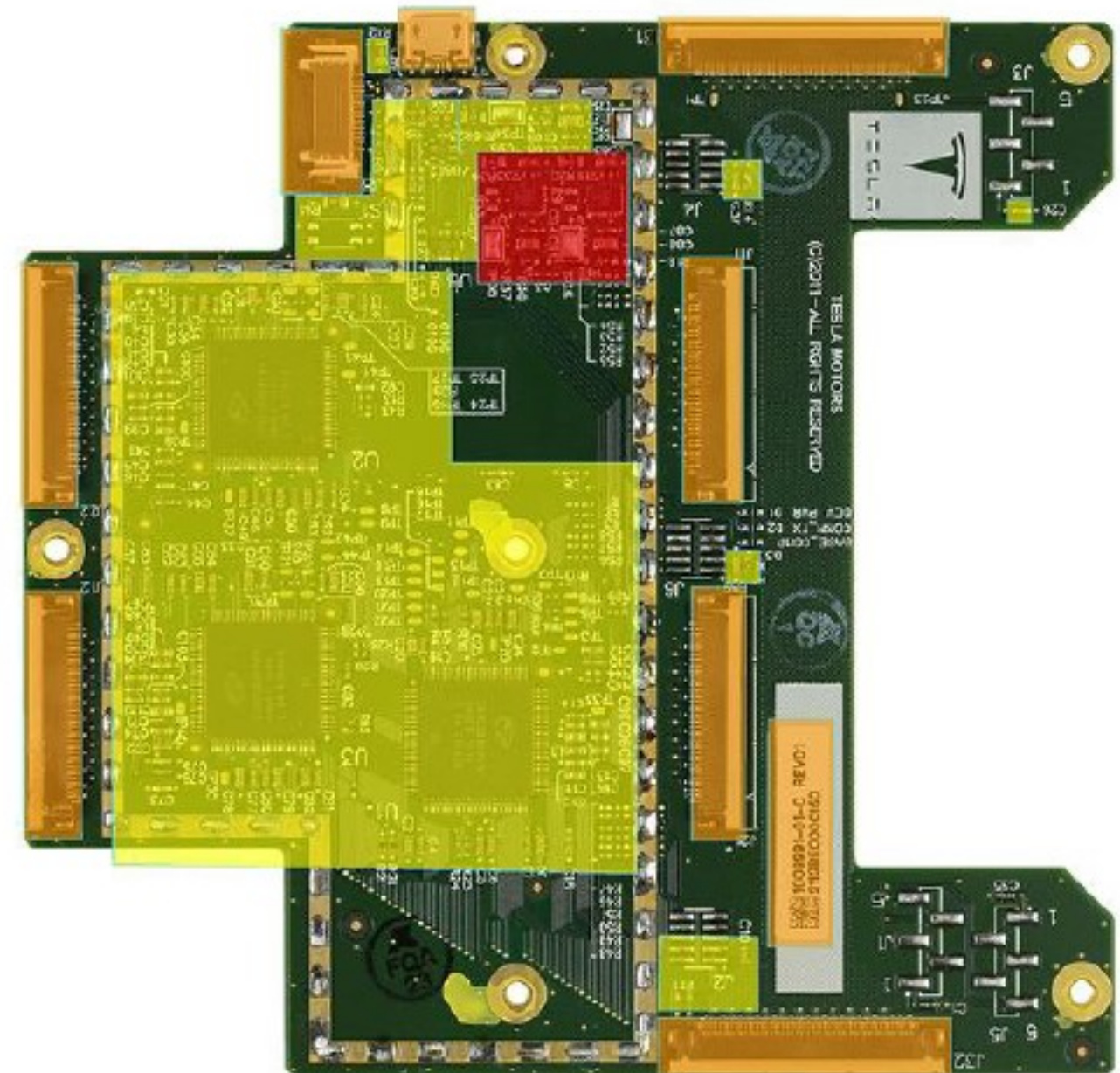
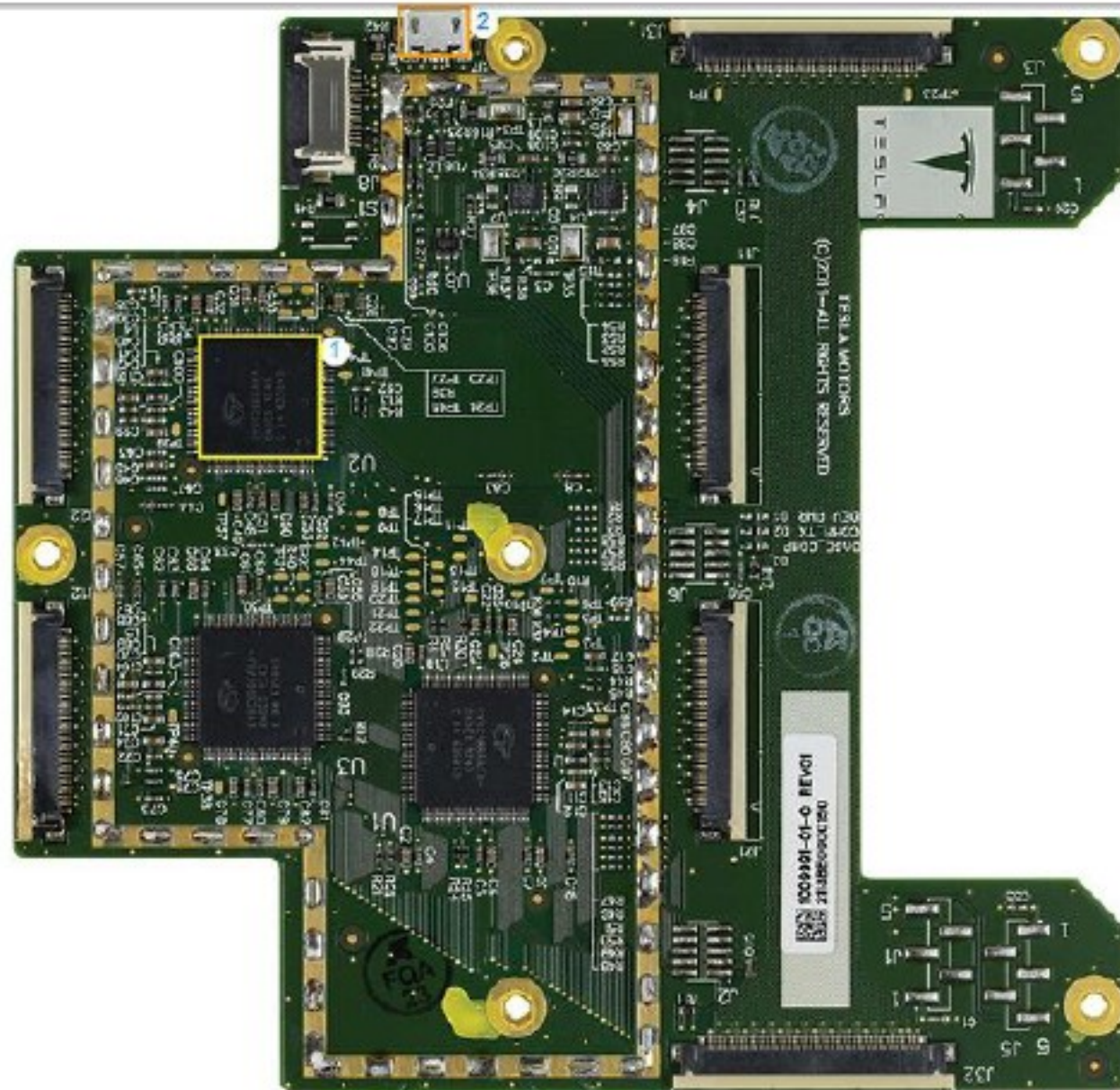
Main PCB, Bluetooth/WLAN Module



Components

1. PARROT, CPICS00251A, Bluetooth Handsfree ASIC, Parrot6i Interface IC
2. TOSHIBA SEMICONDUCTOR, TC58NYG0S3EBAI4, Flash, NAND, 1Gb, SLC
3. SMT Cover, Stamped / Formed Metal
4. MARVELL SEMICONDUCTOR INC, 88W8688-BFL2, Bluetooth / WLAN, SoC, IEEE 802.11 a/b/g, Bluetooth 3.0+HS
5. MICRON TECHNOLOGY INC, MT46H16M16LFBF-6 IT:H, SDRAM, Mobile DDR, 256Mb, 167MHz
6. SMT Frame, Stamped / Formed Metal

Touch Controller PCB - Bottom



Components

- 1. CYPRESS SEMICONDUCTOR CORP,CY8C3866AXA-040ES,Programmable SoC,67MHz 8051 CPU Core,64KB Flash,8KB SRAM,62 GPIOs,w/ Capacitive Sensing,Automotive
- 2. Jack, Micro USB, Right Angle, w/ Metal Housing

About IHS

IHS服务内容

IHS拥有一支由8000名行业信息研究员、分析师、学术权威专家和软件开发人员组成的优秀专家团队，他们对IHS公司所服务的行业拥有深刻的认识和深厚的积淀。

五十多年来，IHS以提供卓越、专业的客户服务为使命，向客户提供大量重要资讯和分析；提供的专业信息解决方案帮助客户规避风险、梳理复杂问题、制订有针对性的策略，最终让他们能够做出更好的商业决策。

信息

分析

专业
知识

IHS服务对象

我们的客户包括几乎所有行业的各级决策者。
无论您是代表跨国企业还是地方政府，
IHS提供的服务都能满足您的需求。

IHS 服务于：

165+
国家所在的企业和政府

客户从小型企业
到
全球500强企业

75+%
全球500强企业

70+%
财富1000强企业

IHS产业覆盖

航空航天与国防

拥有超过100年的丰富经验，能为客户提供关于国防安全装备、市场、行业和风险方面无与伦比的新闻，情报和见解

化学工业

拥有超过200位全球领先的行业权威人士，为300多个化工市场和95种行业的2000个工艺提供综合见解和分析

国家经济与风险

对全球200多个国家和170多个行业的商业环境、经济前景和风险提供深度分析

科技

全球最大的电子零组件数据库，拥有超过3.5亿个零件的信息



汽车

全球最大的汽车分析师团队，在整个汽车价值链拥有数百位专家分布于全球15个关键市场

能源

丰富的石油和天然气井信息可以追溯到1860年，覆盖全球550万口油气井

海事

全球最大的海事数据库，信息收集历史超过250年，提供所有100吨级以上船舶的综合信息

产品设计

为客户提供所需信息，使其能够在从设计、研发到生产、维护最终换代的全生命周期进行产品管理。同时也让企业能够获得相应阶段的规范和标准。

环境、健康、安全与可持续性

为客户提供所需解决方案，帮助他们做出与环境、健康和安全风险、产品监管、温室气体和企业社会责任有关的关键决策。